

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7				General trend A1 {At lower concentrations / or given values / dilute solutions} mass increases / ORA A2 water moves in because the {water potential of solution outside cells is higher than inside / correct ref to water potential gradient} so / ORA A3 {the line crosses the axis / value from graph} where there is no change in mass A4 {the water potential of the solution inside and outside the cells is the same / so there is no net movement of water / this is the point of incipient plasmolysis / OWTTE} A5 {At higher concentrations / or given values / concentrated solutions} mass decreases A6 water moves out because the {water potential of solution outside cells is lower than inside / correct ref to water potential gradient} so Reject reference to water concentration. Difference between potato and sweet potato B1 Sweet potato (cells) contain (more) sugars B2 Sweet potato (cells) have a lower water potential than potato (cells) / ORA B3 The concentration at which there is {no change / owtte} in mass is higher in sweet potato than in potato B4 ref to gain in mass being greater at {0M / lower concentrations} sucrose in sweet potato / ref. to loss in mass less in sweet potato at {1M / higher concentrations (sucrose)} B5 Correct ref to data comparing potato and sweet potato to give evidence for B3 or B4	3	4	2	9		4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				Animal tissue C1 Animal {tissue / cells} would {decrease in mass / lose water / appear crenated} if placed in solution of lower water potential / hypertonic solution. C2 There would be no change in mass if the external solution is isotonic. C3 If the external solution has a {higher water potential / is hypotonic} compared to {tissue/cells}, water would enter the cells. C4 cells would burst. C5 no cell wall.						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				7-9 marks Indicative content of this level is detailed content from all three areas. <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed content from two areas or less detail from three areas. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any correct indicative content. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks The candidate does not make any attempt or give a relevant answer worthy of credit.						
				Question 7 total	3	4	2	9		9